

Dr. Hariom Gangwar
MBBS (KGMU)

NCERT

TABLET

- Flow charts for easy and better understanding
- Short, concise points - easy to remember
- Completely updated NCERT till 2020
- All 38 Chapters Covered



**Latest
Edition
2020**

**Best handbook
for NEET
BIOLOGY**

NCERT Filtrate

Scientists and Related Fields

1. Ernst Mayr (1904-2004)	: Evolutionary Biologist
2. Matthias Schleiden	: German Botanist
3. Julius von Sachs	: German Botanist
4. Theodore Schwann	: British Zoologist
5. Cornelius van Niel	: Microbiologist
6. Alfonso Corti	: Italian Anatomist
7. Dr. W. Dudgeon	: American Missionary Teacher.
8. P. Maheshwari	: Indian Botanist
9. Reginald C. Punnet	: British Geneticist
10. George Gammow	: Physicist
11. Alfred Wallace	: Naturalist
12. Lamarck	: French Naturalist
13. Edward Wilson	: Sociobiologist.
14. Paul Ehrlich	: Stanford Ecologist
15. Ramdeo Misra	: Indian Ecologist
16. Francois Jacob	: Geneticist
17. Jacque Monad	: Biochemist
18. Alexander von Humboldt	: German Naturalist & Geographer
19. Rene Descartes	: French Philosopher, Mathematician and Biologist.

Sizes in NCERT Biology

1. Diameter of pollen grain	: 25-50 micrometer.
2. Diameter of RBC	: 7 micrometer.
3. Smallest cell	: Mycoplasma (0.3 micron in length)
4. Largest cell	: Egg of an ostrich
5. Typical eukaryotic cell	: 10-20 Micron

6. Typical eukaryotic plant cell : 50 micrometer
7. Typical plant cell : Onion cell
8. Typical eukaryotic cell cycle : Human cells in culture.
9. Size of bacteria : 3-5 micron
10. Typical bacteria : 1-2 micrometer
11. Viruses : 0.02 - 0.2 micrometer
12. Diameter of G.B. cisternae : 0.5-1 Micron
13. Diameter of mitochondria : 0.2-1 μm (Average 0.5 μm)
14. Length of mitochondria : 1-4.1 micron
15. Length of chloroplast : 5-10 micrometer (μm)
16. Width of chloroplast : 2-4 micrometer
17. Length of *E. coli* DNA : 1.36 mm
18. Length of Human DNA : 2.2 m

Experimental Materials

1. Engleman experiment : *Cladophora* (green algae).
2. Priestely experiment : Mint plant
3. Van Niel experiment : Purple & green sulphur bacteria
4. Mendel's work : Garden pea (*Pisum sativum*).
5. Hugo de Vries work on mutation : Evening primrose.
6. Morgan's work : *Drosophila melanogaster*.
7. Griffith experiment : *Streptococcus pneumoniae*.
8. Hershy & Chase experiment : Bacteriophage [Bacterial viruses]
9. Meselson & Stahl experiment : *E. coli* bacteria.
10. Taylor's work : *Vicia faba* (faba beans)
11. Melvin Calvin's work : Algal photosynthesis.

Ph.D.

1. 1931 : Katherine Esau (United state).
2. 1937 : Ramdeo Misra - In Ecology (From Leeds University, U.K.)
3. 1949 : G.N. Ramachanadaran (From Cambridge university)
4. 1950 : Watson-The effect of hard X-rays on bacteriophage multiplication
5. 1954 : Crick - X-ray diffraction: polypeptides & protein.
6. 1935 : Melvin Calvin-Ph.D. in chemistry (from University of Minnesota).

NOBEL PRIZES

1. 1945 : Fleming, Chain & Florey.

2. 1961 : Calvin (in Chemistry)
3. 1962 : Watson, Crick & Wilkins.
4. 1968 : Holley, Nirenberg & Hargobind Khorana.
5. 2005 : Chavin, Grubbs & Shrock (in chemistry).

OTHER PRIZES

1. Padma Bhushan Award : M.S. Swaminathan
2. Sanjay Gandhi Award in Environment and Ecology : Ramdeo Misra.
3. Triple Crown (Balzan Prize in 1983, the international prize for biology in 1994, and the Crafoord prize in 1999) : Ernst Mayr.
4. Lasker Award : Watson & Crick.

Term coined by

1. Term Virus : **Ivanowsky (NCERT Correction-2019-20)**
2. Term Chromatin : Flemming
3. Term Linkage : Morgan
4. Term Recombination : Morgan
5. Term Mutation : Hugo de Vries
6. Term Biodiversity popularised by : Edward Wilson.

Units

1. Unit of classification : Taxon/category/rank.
2. Unit of Neural system : Neurons
3. Functional unit of kidney : Nephron.
4. Anatomical unit of muscle : Muscle fibre/muscle cell.
5. Functional unit of contraction : Sarcomere
6. Functional unit of inheritance : Genes.
7. Unit of Noise Pollution : dB (Decibel).
8. Unit of ozone layer thickness : DU (Dobson Units).
9. Functional unit of Nature : Ecosystem.
10. Structural and functional unit of all living organisms : Cell.
11. Reproductive unit in the angiosperms : Flower
12. Unit of water potential : Pascals (Pa) or any other pressure unit.
13. Structural and functional units of liver : Hepatic lobules
14. Structural and functional unit between developing embryo (foetus) and maternal body : Placenta
15. Functional unit of kidney : Nephron

Data of NCERT for Humans

1. Weight of Liver : 1.2 to 1.5 kg
2. Weight of Kidney : 120 to 170 gm
3. Number of lobes in Liver : 2
4. Number of salivary glands : 3 pairs
5. Number of portions in stomach : 4 (Cardiac, fundic, body & Pyloric).
6. Stomach stores food for : 4 to 5 hours.
7. pH of saliva : 6.8
8. pH of gastric juice : 1.8
9. pH in duodenum : 7.8
10. pH of urine : 6.0
11. Breathing rate : 12-16 times/minute
12. Tidal volume : 500 mL
13. IRV : 2500 - 3000 mL
14. ERV : 1000 - 1100 mL
15. RV : 1100 - 1200 mL
16. Water in plasma : 90-92%
17. Water in watermelon : 92%.
18. Water in herbaceous plants : 85-90%.
19. Water in human cell : 70-90%.
20. Colon is divided in : 4 parts (Ascending, transverse, descending and sigmoid parts). (**NCERT Correction-2019-20**)
21. Solubility of CO_2 is 20-25 times higher than that of O_2 .
22. Diffusion membrane is made up of 3 layers.
23. Total thickness of diffusion membrane : < 1 mm
24. No. of wall layers in alimentary canal (from oesophagus to rectum) : 4
25. Gastric gland have 3 type of cells : Mucus neck cells, chief cells and parietal cells
26. Every 100 ml of oxygenated blood can deliver how much O_2 to tissues : 5 mL
27. Every 100 ml of deoxygenated blood delivers how much CO_2 to the alveoli : 4 mL

Shapes in NCERT Biology

1. Pollen grains are generally : Spherical
2. Nucleolus : Spherical
3. Centriole : Cylindrical

4. Mitochondria	: Sausage shaped or cylindrical
5. RBCs	: Round & biconcave
6. Mesophyll cell	: Round & Oval
7. Trachied	: Elongated
8. Columnar epithelium	: Long & narrow.
9. WBCs	: Amoeboid.
10. Nerve cell (longest cell)	: Long & Branched
11. Henle's loop	: Hairpin bend
12. Stomach	: J-shaped
13. Spleen & kidney	: Bean shaped
14. Thymus	: Lobed organ.
15. Patella	: Cup shaped
16. Bowman's capsule	: Double walled cup-like structure
17. Heart	: Clenched fist (size)
18. Oxygen dissociation curve	: Sigmoid.
19. Diaphragm	: Dome shaped.
20. Smooth muscle	: Fusiform.
21. Setae	: S-shaped
22. Intestinal caecae	: Conical
23. Root hair	: Thread like
24. Root cap	: Thimble like
25. Tongue of frog	: Bilobed
26. Nucleus of vegetative cells	: Irregular
27. Generative cell	: Spindle shaped.
28. Ommatidia of cockroach	: Hexagonal
29. Anther	: Four sided (Tetragonal)
30. Cotyledon (maize /grass)	: Shield shaped.
31. Archegonium of bryophytes	: Flask shaped.
32. Gills of molluscs	: Feather like.
33. Radula of molluscs	: File like
34. Chiasmata	: X- shaped
35. Testis	: Oval
36. Uterus	: Inverted pear.
37. Fimbriae & Clitoris	: Finger like.

38. Vasa recta & hyoid bone : U-shaped
39. Duodenum : C-shaped
40. Verhulst - Pearl logistic growth curve : Sigmoid
41. Exponential or geometric growth curve : J - shaped
42. In female cockroach, 7th sternum is : Boat shaped
43. Antennae & anal styles of cockroach : Thread like
44. Head of cockroach, Scapula and Sinus venosus of frog : Triangular
45. Microsporangium : Near circular in outline (in transverse section).
46. Male gonapophysis or phallomere : Asymmetrical structure (Chitinous)

FULL FORMS

1. ICBN : International Code for Botanical Nomenclature
2. ICZN : International Code of Zoological Nomenclature
3. NBRI : National Botanical Research Institute (Lko.)
4. CDRI : Central Drug Research Institute (Lko.)
5. IBG : Indian Botanical Garden (Howrah)
6. PSTD : Potato Spindle Tuber Disease
7. PPLO : Pleuro Pneumonia Like Organisms
8. [S] : Concentration of substrate
9. RRC : Respiratory Rhythm Centre (Medulla)
10. GFR : Glomerular Filtration Rate
11. JGA : Juxta-Glomerular Apparatus
12. ANF : Atrial-Natriuretic factor
13. GnRH : Gonadotrophin Releasing Hormone
14. GH : Growth Hormone
15. PRL : Prolactin
16. TSH : Thyroid Stimulating Hormone
17. ACTH : Adreno Corticotrophic Hormone
18. MSH : Melanocyte Stimulating Hormone
19. ADH : Vasopressin/Anti-diuretic Hormone
20. T₃ : Triiodothyronine
21. T₄ : Tetraiodothyronine or thyroxine
22. TCT : Thyrocalcitonin
23. PTH : Parathyroid hormone
24. CCK : Cholecystokinin
25. GIP : Gastric Inhibitory Peptide

26. LHC : Light Harvesting Complex
27. PGA : Phosphoglyceric acid
28. OAA : Oxalo acetic acid
29. RuBisCO : Ribulose biphosphate carboxylase-oxygenase.
30. NAD : Nicotinamide Adenine Dinucleotide (Coenzyme).
31. CoA : Coenzyme A
32. TCA : Tricarboxylic Acid Cycle or Citric Acid Cycle.
33. ETC : Electron Transport Chain
34. RQ : Respiratory Quotient
35. PGRs : Plant Growth Regulators
36. NAA : Napthalene Acetic Acid
37. 2, 4-D : 2, 4-Dichlorophenoxy acetic acid
38. GA₃ : Gibberellic acid.
39. PMC : Pollen mother cell/microscope mother cell.
40. MMC : Megaspore mother cell.
41. PEN : Primary Endosperm Nucleus.
42. hCG : Human Chorionic Gonadotrophin
43. hPL : Human Placental Lactogen
44. RCH : Reproductive & Child Health Care.
45. STDs : Sexually Transmitted Diseases
46. VD : Venereal disease.
47. RTI : Reproductive Tract Infections
48. WHO : World Health Organisation
49. MMR : Maternal Mortality Rate
50. IMR : Infant Mortality Rate
51. IUDs : Intra Uterine Devices
52. OCPs : Oral Contraceptive Pills
53. MTP : Medical Termination of Pregnancy
54. ART : Assisted Reproductive Technologies
55. ZIFT : Zygote Intra-fallopian transfer
56. GIFT : Gamete Intra-fallopian transfer
57. IUT : Intra Uterine Transfer
58. IUI : Intra Uterine Insemination
59. AIT/AI : Artificial Insemination Technique

60. ICSI	: Intra cytoplasmic sperm injection
61. IVF	: In-vitro fertilisation
62. ET	: Embryo transfer
63. SCA	: Sickle cell Anaemia
64. PKU	: Phenylketonuria
65. NHC	: Non-histone Chromosomal proteins
66. snRNA	: Small Nuclear RNA
67. hnRNA	: Heterogenous Nuclear RNA
68. sRNA	: Soluble RNA
69. UTR	: Untranslated Regions
70. HGP	: Human Genome Project
71. ELSI	: Ethical, Legal & Social Issues
72. BAC	: Bacterial Artificial Chromosome
73. YAC	: Yeast Artificial Chromosome
74. ESTs	: Expressed Sequence Tags
75. SNPs	: Single Nucleotide Polymorphism
76. VNTR	: Variable Number of Tandem Repeats
77. SCRR	: School of Cytogenetics & Radiation Reserch
78. IARI	: Indian Agricultural Research Institute (New Delhi)
79. CMI	: Cell - Mediated Immunity
80. MALT	: Mucosal Associated Lymphoid Tissue
81. AIDS	: Acquired Immuno Deficiency Syndrome
82. HIV	: Human Immuno Deficiency Virus
83. ELISA	: Enzyme Linked Immuno Sorbent Assay
84. NACO	: National AIDS Control Organisation
85. NGOs	: Non-Governmental Organisation
86. C-onc	: Cellular Oncogenes
87. CT	: Computed Tomography
88. MRI	: Magnetic Resonance Imaging
89. LSD	: Lysergic acid diethyl amide
90. MOET	: Multiple Ovulation Embryo Transfer Technology.
91. ICWMI	: International Centre for Wheat & Maize Improvement (Mexico)
92. IRRI	: International Rice Research Institute (Philippines)

- 93. SCP : Single Cell Protein
- 94. LAB : Lactic Acid Bacteria
- 95. STPs : Sewage Treatment Plants
- 96. BOD : Biochemical oxygen Demand
- 97. DO : Dissolved Oxygen
- 98. KVIC : Khadi & Village Industries Commission
- 99. Bt : *Bacillus thuringiensis*.
- 100. NPV : Nucleopolyhedro virus
- 101. IPM : Integrated Pest Management
- 102. EcoRI : *Escherichia coli* RY13.
- 103. Ori : Origin of Replication
- 104. pBR322 : Plasmid
- 105. Ti Plasmid : Tumor Inducing Plasmid
- 106. Taq : *Thermus aquaticus*
- 107. PCR : Polymerase Chain Reaction
- 108. GMO : Genetically modified Organisms
- 109. RNAi : RNA interference
- 110. ADA : Adenosine deaminase
- 111. GEAC : Genetic Engineering Approval Committee.
- 112. NCEPC : National Committee for Environmental Planning & Coordination (1972)
- 113. MOEF : Ministry of Environment & Forests (1984)
- 114. INSA : Indian National Science Academy
- 115. NPP : Net Primary Productivity
- 116. GPP : Gross Primary Productivity
- 117. GNP : Gross National Product
- 118. GDP : Gross Domestic Product
- 119. GFC : Grazing food chain
- 120. DFC : Detritus food chain
- 121. IUCN : International Union for Conservation of Nature & Natural Resources
- 122. CPCB : Central Pollution Control Board
- 123. CNG : Compressed Natural Gas
- 124. ppb : Parts per billion
- 125. ppm : Parts per million

- 126. FOAM : Friends of Arcata Marsh
- 127. DU : Dobson Units
- 128. dB : Decibel
- 129. CFCs : Chlorofluorocarbons
- 130. JFM : Joint Forest Management (1980).

Important Cells in NCERT

1. **Heterocysts** : Specialised cells found in **Nostoc** & **Anabaena** which help in fixation of atmospheric nitrogen.
2. Smallest living cell is **Mycoplasma**
3. **Choanocytes** or collar cells - Found in sponges that line the spongocoel & canals.
4. **Cnidoblasts**/cnidocytes - Found in coelentrata/cnidaria which contain the stinging capsules or nematocytes.
5. **Flame cells**- Found in flat worms and help in osmoregulation & excretion.
6. **Statocysts**- These are balancing organs found in arthropods.
7. Root hair- Found in zone of maturation and help in water & mineral absorption.
8. Guard cell- Encloses stomatal aperture or pore. Bean-shaped but in grasses dumb-bell shaped.
9. Subsidiary cells- A few epidermal cells, in the vicinity of the guard cells become specialised in their shape & size & called subsidiary cells
10. Bundle-sheath cells- The vascular bundles are surrounded by a layer of thick walled bundle-sheath cells.
11. **Bulliform cells**- In grasses, certain adaxial epidermis cells along the veins modify themselves into large, empty & colourless cells.
12. Chondrocytes- Cells of cartilage
13. Osteocytes- Bone cells.
14. Neuroglial cells- Found in the neural system which protect & support neurons.
15. **Goblet cells**- Present in intestinal mucosal epithelium which secrete mucus.
16. Mucus neck cells- Present in gastric gland & secrete mucus
17. Peptic/cheif cells- Present in gastric gland & secrete pepsinogen.
18. Parietal/oxyntic cells- Present in gastric gland & secrete HCl & intrinsic factor.

19. Hepatic cells or hepatocytes- Found in liver.
20. **JG (Juxtaglomerular) cells**- Release **renin** & **erythropoietin** hormones
21. Neurosecretory cells (Nuclei) found in hypothalamus, which secrete hormones.
22. Ganglion cells- Found in retina
23. Bipolar cells- Found in retina
24. Photoreceptor cells- Found in retina
25. Rod cells- For twilight (scotopic vision)
26. Cone cells- For day light (photopic vision)
27. Organ of Corti contains hair cells
28. Adipocytes- Cells of adipose tissue
29. **PMNL** (Polymorpho Nuclear Leukocytes)- A type of neutrophil providing cellular barrier in innate immunity.
30. **Natural killer cells**- Type of lymphocyte providing cellular barrier.
31. Macrophages- Present in tissue (areolar tissue) that can phagocytose & destroy microbes providing cellular barrier.
32. Inner cell mass of human blastocyst contains **stem cells** which have the potency to give rise to all the tissues & organs.
33. The epithelial cells of Bowman's capsule called **podocytes**.

Contributions of Biologists

1. Binomial nomenclature was given by **Carolus Linnaeus**.
2. Two kingdom system of classification was given by Carolus Linnaeus.
3. Systema Naturae was written by Carolus Linnaeus.
4. Artificial system of classification was given by Carolus Linnaeus.
5. Natural System of classification for flowering plants was given by **George Bentham** & **J.D. Hooker**.
6. 5 kingdom system of classification was given by **R.H. Whittaker** (1969).
7. **Aristotle** was the earliest to attempt a more scientific basis for classification.
8. Biological classification of plants & animals was first proposed by Aristotle.
9. Causal organism of the mosaic disease of tobacco is recognised by D.J. Ivanowsky (1892).
10. Contagium vivum fluidum is called by M.W. Beijerinck (1898).
11. Viruses crystallised by W.M. Stanley (1935).

12. Viroids were discovered by T.O. Diener (1971).
13. 'Anatomy of Seed Plants' published by **Katherine Esau** (1960).
14. Curly top virus spreads through a plant via phloem tissue, it is reported by Katherine Esau
15. Founder of Madras School of Conformational analysis of biopolymers was **G.N. Ramachandran**.
16. Triple helical model of collagen was given by G.N. Ramachandran
17. Ramachandran was influenced by Linus Pauling
18. Darwin was influenced by Malthus
19. P. Maheshwari was influenced by Dr. W. Dudgeon
20. α - helix & β - sheet structures was published by **Linus Pauling**
21. **Anton von Leeuwenhoek** first saw & described a live cell.
22. Cell theory was given by Schleiden & Schwann.
23. Cell theory was modified by Rudolf Virchow (1855)
24. Fluid mosaic model was given by Singer & Nicolson (1972)
25. Golgi bodies was discovered by Camillo Golgi (1898)
26. Ribosome was discovered by **George Palade** (1953)
27. Nucleus was first discovered and described by Robert Brown (1831)
28. Melvin Calvin earn Nobel Prize in 1961 for mapping of the pathway of carbon assimilation in PHS.
29. Calvin work with J.A. Bassham
30. **Hydroponics** technique developed by **Julius von Sachs** (1860).
31. O_2 was discovered by Joseph Priestley (1774)
32. Priestley revealed the essential role of air in growth of green plants (1770).
33. **Jan Ingenhousz** used a similar setup as the one used by Priestley.
34. Jan Ingenhousz showed that sunlight is essential to the plant process.
35. Jan Ingenhousz showed that it is only the green part of the plants that could release oxygen.
36. Julius von Sachs provide evidence for production of glucose when plants grow.
37. Sachs shows that the green substance in plants (chlorophyll) is located in special bodies (later called chloroplasts) within plant cells.
38. Sachs found that the green parts in plants is where glucose is made & that the glucose is usually stored as starch.
39. **First action spectrum** of PHS was described by **T.W. Englemann**.

40. Van Niel demonstrated that PHS is essentially a light-dependent reaction in which hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrate.
41. **Van Niel** inferred that the O_2 evolved by green plant comes from H_2O , not from CO_2 .
42. Calvin use radioactive C^{14} in algal photosynthesis studies that led the discovery that the first CO_2 fixation product was a 3-carbon organic acid.
43. Whole pathway of C_3 - cycle was worked by Calvin and Benson.
44. C_4 - pathway was discovered by Hatch & Slack
45. **Law of limiting factors** was given by **Blackmann** (1905)
46. Scheme of Glycolysis was given by Embden, Meyerhof & Parnas. (EMP)
47. Discovery of PGRs is started with the observation of Charles Darwin & his son Francis Darwin
48. Charles Darwin & Francis Darwin observed that the coleoptiles of canary grass respond to unilateral illumination by growing towards the light source (Phototropism).
49. **Auxin** was isolated from tips of coleoptiles of oat seedlings by **F.W. Went**.
50. **E. Kurosawa** reported the appearance of symptoms of "**Bakane (foolish seedlings) disease**" in uninfected rice seedlings when they were treated with sterile filtrates of the fungus **Gibberella fujikuroi**.
51. Skoog & his Co-workers observed that from the internodal segments of tobacco stems the callus proliferate only if, in addition to auxins, the nutrient medium was supplemented with one of the following : extract of vascular tissue, Yeast Extract, Coconut milk or DNA.
52. **Skoog & Miller** identified, crystallised & termed Kinetin.
53. Cousins confirmed the release of volatile substance from ripened oranges that hastened the ripening of stored unripened bananas.
54. P. Maheshwari established department of botany at university of Delhi as important centre of research in Embryology & Tissue culture.
55. P. Maheshwari emphasised the need for initiation of work on artificial culture of immature embryos.
56. P. Maheshwari work on test-tube fertilisation & intra-ovarian pollination
57. P. Maheshwari popularised the use of embryological character in taxonomy.
58. **P. Maheshwari & Ramdeo Misra** was awarded by **INSA** (Indian National Science Academy).
59. Sanjay Gandhi Award in Environment & Ecology is given to Ramdeo Misra.

60. **M.S. Swaminathan** is awarded by **Padma Bhushan**.
61. **Lasker award** given to **Watson & Crick**
62. **Balzen prize** given to **Ernst Mayr**
63. P.Maheshwari was honoured with Fellowship of Royal Society of London (FRS).
64. Short-duration high-yielding varieties of rice (including scented Basmati) was developed by M.S. Swaminathan.
65. M.S. Swaminathan initiated collaboration with Norman Borlaug
66. M.S. Swaminathan is initiator of Lab-to-land & food security.
67. M.S. Swaminathan is known for the development of the concept of crop cafeteria & crop scheduling.
68. Good humor hypothesis was given by Hippocrates & Indian Ayurveda System
69. Good humor hypothesis was disproved by William Harvey.
70. **Blood circulation** was discovered by - **William Harvey**
71. **Rivet popper hypothesis** was given by **Paul Ehrlich** (Stanford Ecologist)
72. **Polyblend** is formed by **Ahmed Khan** & his company.
73. Ramesh Chandra Dagar is related to organic farming.
74. Semi-dwarf wheat at ICWMI were developed by **Norman Borlaug**.
75. **Herbert Boyer** observed that the restriction enzymes have the capability of cutting DNA strands in a particular fashion (1969).
76. Stanley Cohen developed a method of removing plasmids from the cell & then reinserting them in other cells.
77. First recombinant DNA was formed by Cohen & Boyer.
78. Watson study on the "Effect of hard X-rays on bacteriophage multiplication" (1950)
79. Crick study on "X-ray diffraction: polypeptides & proteins" (1954)
80. **Double helical model** of B-DNA was given by **Watson & Crick** (1953)
81. Mendal proposed laws of inheritance in living organisms.
82. Punnet square were developed by R.C. Punnet.
83. Mendel's results were rediscovered by de Vries, Correns & von Tschermak independently in 1900.
84. Sutton & Boveri noted that the behaviour of chromosome was parallel to the behaviour of genes.
85. **Chromosomal theory of inheritance** was given by **Sutton & Boveri**.
86. Experimental verification of the chromosomal theory of inheritance was given by **T.H. Morgan**

87. Sutton & Boveri argued that the pairing & separation of a pair of chromosomes would lead to the segregation of a pair of factors they carried.
88. **Genetic map** discovered by **Alfred Sturtevent**.
89. Henking trace a specific nuclear structure (X-body) all through spermatogenesis in a few insects (1891).
90. Down's syndrome was first described by Langdon Down (1866)
91. **DNA** (Nuclein) was first identified by **Friedrich Meischer** (1869).
92. **X-ray diffraction** data produced by **Wilkins & Franklin**.
93. **Transforming principle** was given by **Griffith** (1928).
94. Biochemical characterisation of transforming principle was given by Avery, MacLeod & McCarty.
95. Unequivocal proof that DNA is the genetic material came from the experiments of **Hershey & Chase** (1952).
96. **Replication scheme** was given by **Watson & Crick**.
97. **Central dogma** of biology was given by **Crick**.
98. The experimental proof that DNA replicates semiconservatively was given in prokaryotes by **Meselson & Stahl** (1958).
99. The experimental proof that DNA replicates semiconservatively was given in eukaryotes by **Taylor** (1958).
100. George Gamow (a physicist) proposed that codon was triplet.
101. **Har Gobind Khorana** synthesizes RNA molecules with defined combination of bases (homopolymers & copolymers).
102. Nirenberg in cell-free system for protein synthesis finally helped the code to be deciphered.
103. Polynucleotide phosphorylase is called Severo Ochoa enzyme
104. **Lac operon** was given by **Jacob & Monod**.
105. Automated DNA sequencers worked on the principle of a method developed by Frederick Sanger
106. **Frederick Sanger** developed the method for determination of amino acid sequences in proteins.
107. **DNA fingerprinting** technique was developed by **Alec Jeffreys**.
108. Spontaneous generation theory was dismissed by Louis Pasteur.
109. Life comes from pre-existing life it is told by Louis Pasteur.
110. **Oparin & Haldane** proposed that the first form of life could have come from pre-existing non-living organic molecules.

111. **Darwin** made a sea voyage in a sail ship (**H.M.S. Beagle**).
112. Alfred Wallace worked in Malay Archipelago.
113. 'Use and disuse of organs' theory of evolution was given by Lamarck (French Naturalist).
114. Malthus work on population.
115. **Hugo de Vries** brought forth the idea of **mutations**.
116. Hugo de Vries used the term saltation.
117. Ramdeo Misra revered as the father of ecology in India.
118. Ramdeo Misra formulated the first postgraduate course in ecology in India.
119. "Competitive Exclusion principle" was given by Gause.
120. **Resource partitioning** was given by **Mac Arthur**.
121. Cost of ecosystem services was given by **Robert Costanza**.
122. Term Biodiversity popularised by Edward Wilson (Sociobiologist)
123. **Species-area relationship** was given by **Alexander von Humboldt**.
124. The importance of species diversity to the ecosystem was given by David Tilman

Dependency in NCERT Biology

1. Algae are classified into 3 classes, depending on the **type of pigment** and the type of stored food.
2. Kingdom fungi is divided into various classes based on morphology of the mycelium, mode of spore formation & fruiting bodies.
3. Simple epithelium is divided into 3 types (squamous, cuboidal and columnar) on the basis of structural modifications of the cells.
4. **Pollination** can be divided into 3 types (autogamy, geitonogamy & xenogamy) depending on the **source** of pollen.
5. Respiratory quotient depends upon the type of respiratory substrate used during respiration.
6. Neurons can be divided into 3 types (multipolar, bipolar & unipolar) based on the number of axon & dendrites.
7. **Muscles** are of 3 types (skeletal, visceral & cardiac) based on their **location**.
8. Muscles are classified as red & white fibres based primarily on the amount of red coloured myoglobin pigment in them.
9. Methods of locomotion performed by animals vary with their habitat & the demand of the situation.

10. Nature of nitrogenous wastes formed & their excretion vary among animals, mainly depending on the habitat (availability of water).
11. **ABO** blood grouping is based on the presence or absence of the 2 surface **antigens** on the RBCs namely A & B.
12. Mechanism of breathing vary among different groups of animals depending mainly on their habitats & level of organisation.
13. Chromosomes can be classified into 4 types (metacentric, sub-metacentric, acrocentric & telocentric) based on the position of the centromere or 1^o constriction.
14. Enzymes are classified into 6 classes based on the type of reactions they catalyse.
15. Prophase - I is divided into 5 phases (**Code : L郑DD Lata Zara Pani De Do**) based on chromosomal behaviour.
16. Membrane protein can be classified as intergral or peripheral depending on the ease of extraction.
17. Succession of plants is called hydrarch or xerarch, based on the nature of the habitat.
18. Nature & properties of soil is dependent on the climate, weathering process, soil is transported or sedimentary & how soil development occurred.
19. Primary productivity depends on the : (i) Plant species inhabiting a particular area, (ii) Variety of environmental factors, (iii) Availability of nutrients and (iv) Photosynthetic capacity of plants.
20. The rate of decomposition is controlled by chemical composition of detritus & climatic factors (temperature & soil moisture)
21. Organism occupy a specific place in the food chain or a trophic level, based on source of their nutrition or food.
22. In 2^o succession the species that invade depends on the : (i) Condition of soil, (ii) Availability of water, (iii) Environment and (iv) Seed or other propagules present.
23. In dairy form, milk yield is primarily dependent on the quality of breeds in the farm.
24. Different type of alcoholic drinks are obtained depending on the :
 - (i) Type of raw material used for fermentation.
 - (ii) Type of processing (with or without distillation).

Cities and Countries in NCERT

1. Famous Royal Botanical Garden : Kew (England)
2. Indian Botanical Garden : Howrah (India)
3. National Botanical Research Institute (**NBRI**) : Lucknow
4. Central Drug Research Institute (**CDRI**) : Lucknow
5. ***Strobilanthes kunthiana*** found in : Kerela, Karnataka & Tamilnadu.
6. ***Lupinus arcticus*** (Lupine) excavated from : Arctic Tundra
7. ***Phoenix dactylifera*** (Date palm) excavated from : King Herod's Palace near Dead Sea.
8. **Sahiwal Cow** bred in : **Punjab** (through artificial selection & domestication)
9. **Hisardale** (Sheep) develop in : **Punjab** (by cross -breeding)
10. Oparin : Russia
11. Haldane : England
12. S.L. Miller : America
13. Wallace Worked in : Malay Archipelago
14. Observation supporting evolution by natural selection comes from : England
15. Darwin worked in : Galapagos Islands
16. Coelcanth (fish) caught in : South Africa (1938).
17. Few fossils of man-like bones have been discovered in : Ethiopia & Tanzania.
18. About 3-4 mya man - like primate walked in : Eastern Africa
19. *Australopithecus* lived in : East African Grasslands.
20. Fossils of ***Homo erectus*** discovered in : **Java** (1891)
21. Neanderthal man lived in : Near East & Central Asia
22. ***Homo sapiens*** Arose in : **Africa**
23. ***Erythroxylum coca*** is native of : **South America**
24. > 70% live stock population is in : India & China
25. ***Saccharum barberi*** grown in : **North India**
26. ***Saccharum officinarum*** grown in : **South India**
27. IARI : New Delhi
28. IRRI : Phillippines
29. ICWMI : Mexico
30. Taichung Native-I develops from : Taiwan.

31. IR-8 developed from : IRRI, Philippines
32. Borlaug developed semi-dwarf wheat at : ICWMI, Mexico.
33. **Jaya & Ratna** developed in : **India**
34. Toddy is used in : South India
35. Penicillin was used during world war-II to treat : American soldiers
36. **Eli lily** is an : American company.
37. A company which got patent rights on basmati rice : American Company
38. Mango tree do not & can not grow in : Canada & Germany
39. Snow leopard are not found in : Kerela forests
40. High altitude places (>3500 m) : Rohtang pass near Manali & Mansarovar (in China occupied Tibet)
41. In 1920's prickly pear cactus introduced in : Australia
42. Visiting flamingoes & resident fishes compete in : South American lakes
43. **Abingdon tortoise** were found in : **Galapagos Island.**
44. Barnacle **Balanus** & **Chathamalus** found in : Rocky Sea Coasts of Scotland
45. **Rauwolfia vomitoria** growing in : Himalaya
46. Amazonian Rain Forest found in : South America
47. Alexander von Humboldt explored in : South America Jungles
48. **Dodo** extinct from : **Mauritius.**
49. **Quagga** extinct from : **Africa**
50. **Thylacine** extinct from : **Australia**
51. Stellar's sea cow extinct from : Russia
52. Lake Victoria found in : East Africa
53. **Clarius geriepinus** is **African catfish**
54. **Earth Summit** held in : **Rio de Janeiro**
55. **World Summit** held in : **Johannesburg (South Africa)**
56. Montreal Protocol held in : Montreal (Canada)
57. Khasi & Jaintia hills in : Meghalaya
58. Aravalli hills in : Rajasthan
59. Western Ghat regions are found in : Karnataka & Maharastra
60. Sarguja, Chanda & Bastar areas are found in : Madhya Pradesh
61. Ahmed Khan belongs to : Bangalore
62. Ramesh Chandra Dagar related to : Sonipat (Haryana)

63. Ozone hole is particularly marked over : Antarctic region
64. **Slash & Burn cultivation** or Jhum cultivation done in : **North-eastern states** of India.
65. Amrita Devi Bishnoi related to : Jodhpur (Rajasthan)
66. **Chipko movement** started in : **Garhwal Himalayas**
67. **Keolado National Park : Bharatpur (Rajasthan)**
68. Kangaroo rat found in : North American desert.

Years and Contribution of Scientists

1. 1770 : Joseph Priestley - Essential role of air in growth of green plant.
2. 1774 : Joseph Priestley - Discovered O_2 .
3. **1831 : Robert Brown** - 1st discovered and described nucleus.
4. 1838 : Schleiden (German Botanist) - Work on plants.
5. 1839 : Schwann (British Zoologist) - Work on both plants & animals.
6. **1855 : Rudolf Virchow** - **Omnis cellula-e-cellula** (cells arises from pre-existing cells).
7. Life originate from pre-existing life : Pasteur (Yeast).
8. Life originate from non-living (decaying & rotting matter) : Spontaneous generation.
9. Life originate from pre-existing non-living organic molecules (RNA, Protein)- Oparin (Russia) & Haldane (England)- Chemical Evolution.
10. **1856 - 1863 : Mendal experiment** on garden pea. (7 years)
11. 1860 : Julius von Sachs - Developed hydroponics.
12. **1865 : Mendal published his work.**
13. 1866 : Langdon Down - Down's syndrome.
14. 1869 : Friedrich Meischer - Ist identified DNA and named it nuclein.
15. 1891 : Henking - Discovered X-body.
16. 1891 : Fossil discovered in Java - *Homo erectus*.
17. **1892 : Ivanowsky - Discovered virus.**
18. **1898 : Beijerinck - Contagium vivum fluidum** (infectious living fluid).
19. 1898 : Camillo Golgi - Discovered (observed) G.B.
20. 1900 : de Vries, Correns and von Tschermak independently rediscovered Mendel's results.
21. 1902 : Chromosome movement during meiosis has been worked out.
22. **1905 : Law of limiting factor (Blackmann).**

23. 1928 : Fredrick Griffith - Transforming experiment with *Streptococcus pneumoniae* (*Diplococcus*).
24. **1935 : Stanley - Crystallised viruses.**
25. 1937 : Ramdeo Misra obtained Ph.D. in Ecology from Leeds university (U.K.).
26. 1938 : Coelocanth fish caught in South Africa.
27. **1945 : Fleming, Chain & Florey - Awarded Nobel prize.**
28. 1950 : Watson obtained Ph.D. on a study of the effect of hard X-rays on bacteriophage multiplication.
29. **1951 : Family planning programme started in India.**
30. 1952 : Hershey & Chase-experiment on bacteriophage or Bacterial virus gives unequivocal proof that DNA is the genetic material.
31. 1953 : Miller experiment - Methane, ammonia, hydrogen & water vapour.
32. 1953 : Watson & Crick - Double helical structure of B-DNA & replication scheme.
33. 1953 : Palade : Discovered ribosome.
34. 1954 : Ramachandran - Triple helical model of collagen, published in Nature.
35. 1954 : Crick completed Ph.D. on a thesis "X-rays diffraction : polypeptides and proteins.
36. **1958 : Meselson & Stahl** - Work on *E.coli* proves semiconservative replication of DNA in prokaryotes.
37. 1958 : Taylor - Work on *Vicia faba* proves semiconservative replication of DNA in eukaryotes.
38. 1960 : Katherine Esau - Published "Anatomy of Seed Plants."
39. **1961 : Melvin Calvin - Nobel Prize.**
40. 1962 : Watson, Crick and Wilkins - Nobel Prize.
41. 1963 : Wheat varieties (Sonalika & Kalyan sona) introduced.
42. 1963 : Two enzyme responsible for restricting growth of bacteriophage in *E.coli* were isolated.
43. 1966 : Derivative of IR-8 & Taichung native-I introduced.
44. **1969 : Whittaker - Five kingdom classification.**

45. **1971 : Govt. of India legalized MTP.**
46. 1971 : Diener- Discovered Viroid(free RNA without capsid).
47. **1972 : Singer & Nicolson - Fluid mosaic model.**
48. 1972 : Stanley Cohen & Herbert Boyer - Formed 1st recombinant DNA.
49. 1972 : Establishment of NCEPC - National Committee for Environmental Planning & Coordination.
50. **1974 : Water act.**
51. **1980 : Joint forest management, (JFM).**
52. 1981 : AIDS was 1st reported.
53. **1981 : Air act.**
54. 1983 : Eli Lilly (An American company) produces insulin in *E.coli* by recombinant DNA technology.
55. 1984 : Establishment of MOEF : Ministry of Environment & Forest.
56. **1986 : Environment Protection Act.**
57. **1987 : Montreal protocol.** Held in Montreal, Canada.
58. **1987 : Air act amended to include noise as an air pollutant.**
59. **1988 : National Forest Policy.**
60. 1989 : Montreal protocol effective.
61. 1990 : 1st clinical gene therapy.
62. 1990 : HGP launched.
63. 1992 : The Earth summit : Rio de Janeiro.
64. 1997 : 1st transgenic cow- Rosie. It produces human protein enriched milk (2.4 grams per litre). The milk contains the human alpha-lactalbumin
65. 1997 : An American company got patent rights on Basmati Rice.
66. **1997 : Kyoto protocol :** Held in Kyoto(Japan).
67. 2002 : World summit. Held in Johannesburg (South Africa).
68. 2002 : All buses of Delhi were converted by CNG.
69. 2005 : Nobel prize of chemistry given to 'Green chemists' Chauvin, Grubbs & Schrock.